Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/11/2005
PI Name:	Copeland, Robert J. Ph.D.		
Project Title:	A Lightweight Freeze-Tolerant Radiator for an EMU		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced extravehicular activity		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Wheat Ridge	**State:**	CO
Zip Code:	80033-1917	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	08/23/2003	**End Date:**	08/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: PI replaced by James Nabity in 2005.		
Key Personnel Changes/Previous PI:	James Nabity replaces Dr. Copeland as PI; see FY2005 report.		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A new, lightweight, high performance freezable radiator for the Extravehicular Mobility Unit (EMU), has tubes that elastically expand like a spring to accommodate the 9.1% expansion when water freezes. This design maintains all materials within their elastic limits so that relatively large volume changes can be achieved without breaking the tube. We will use all of the available area on an EMU and to reject all of the heat generated in the EMU most of the time and therefore reduce the expendable water required by a factor of ~4 (i.e., only about 25% of the current expendable requirement). We will use advanced carbon composites in our radiator to provide the high thermal conductivity required to transfer the maximum amount of heat to the radiator. Our freezable radiator is therefore very lightweight to respond quickly to changing environments or metabolic loads; freezing and thawing rapidly. Our system not only greatly cuts the use of expendables, but weighs less than the current system. In addition our lightweight, freeze tolerant radiator is scalable to other applications.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The largest and nearest term commercial applications are the use of freeze tolerant tubing on Earth. These Earth-based applications include sprinkler systems and potable water supply in homes and commercial buildings. This market is potentially very large and virtually un-tapped because of the lack of a viable freeze tolerant tube. The Insurance Institute for Property Loss Reduction, says frozen pipes cost the insurance companies in the USA $4.2 billion in damage to insured homes and buildings over the past decade (i.e., about $400,000,000 per year). The savings in insurance rates alone could more than offset the cost to the user, who would have the added benefit of not having non-replaceable photographs and other objects destroyed by water damage and their lives disturbed during repairs of the water damage.
Task Progress:	The development of a freeze tolerant heat rejection system that can accommodate a variable heat load is a key technological innovation of this work. This new thermal rejection system uses a freeze tolerant heat exchanger to transfer heat from the cooling water loop to a loop heat pipe where is rejected by radiation from a lightweight carbon fiber fin. In the first year of this project we have completed initial design of all components of this proposed thermal rejection system for the EMU. We have performed modeling and necessary analysis calculations to size the lightweight carbon fiber radiator fin, the loop heat pipe, and the freeze tolerant heat exchanger. The demonstration of freeze trap tolerance of the heat exchanger is key to the thermal rejection system, as freezing of the cooling loop water in the heat exchanger allows the rate of rejection to be turned down in the low metabolic load condition. We have fabricated a test article of heat exchanger and experimentally verified the effectiveness of the freeze tolerant design. Freeze trap tests on the heat exchanger shown negligible deflection in the metal test article when it underwent multiple freeze/thaw cycles and there was no permanent deformation in the test article. Thus, we have successful demonstrated that Technical Data Analysis (TDA)’s freeze tolerant heat exchanger can be turned down by freezing and can cycle between high and low heat rejection rates.
Bibliography Type:	Description: (Last Updated: 02/27/2004)